

Sentiment Analysis Backends: Comparison (HF vs OpenAI vs Ollama)

This document compares three sentiment-analysis backends used in the project: Hugging Face, OpenAI, and Ollama. The goal is to understand how they differ in speed, accuracy, setup, and prediction method.

Aspect	Hugging Face (HF)	OpenAI	Ollama
Where model runs	Local (CPU/GPU)	Cloud	Local (Ollama runtime)
Model type	5-class DistilBERT classifier	General LLM	Local LLM
How predictions work (part 1)	Uses a trained classifier head	Uses a few-shot prompt	Uses a few-shot prompt
How predictions work (part 2)	Outputs logits → argmax → label	Returns text → parsed into 0–4	Returns text → parsed into 0–4
API key required	No	Yes	No
Internet needed	No	Yes	No
Speed	Fast / predictable	Medium (network)	Slower unless strong hardware
Cost	Free	Paid per token	Free
Accuracy (expected)	Good	Very good	Medium–Good
Determinism	High	High (temp=0)	Good
Memory usage	Loads full model locally	Minimal local usage	Local model loaded in RAM
Fine-tuning	Supported	Supported	Not available
Best for	Reliable and fast local runs	Highest reasoning ability	Offline / privacy
Worst for	Ambiguous reasoning tasks	Offline usage	Weak hardware

Summary

• **Hugging Face:** Fast, local, and accurate for 5-class sentiment. • **OpenAI:** Strongest accuracy overall thanks to LLM reasoning. • **Ollama:** Fully offline alternative; accuracy depends on hardware/model.